

Memorize and Focus on Fine-Grained Objects for Robotic Imitation Learning in Familiar Scene

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Abstract—Imitation learning of visuomotor policies endows the robot with dexterous manipulation skills. However, even in a familiar scene where a robot repeats its tasks, distinguishing a target fine-grained object (FO) from similar objects in the same coarse category remains an unresolved issue. Besides, current visuomotor policies encode the scene image as the visual condition, which might be prone to visual distractors. To address these problems, we first propose FO Memory-enhanced Segment Anything Model 3 (FOM-SAM3). With SAM3 fully frozen, it converts a coarse concept prompt into reusable FO Memory Tokens from a few multi-view registration images. At deployment, the tokens directly prompt frozen SAM3 without registration images or model adaptation. Second, we introduce Focused Spatial-Appearance Encoding (FSAE), which focuses the action policy on ordered target and manipulation-context regions and models their local appearance and relative geometry. FSAE produces a compact condition vector for DP or object condition tokens for ACT without changing either action objective. Experiments on 30 FOs from 4 coarse categories show that FO Memory improves target segmentation, ranks the queried FO above visually similar candidates, suppresses Similar-FO confusion, and rejects absent identities more reliably. Real-robot experiments report completed and failed rollouts and evaluate within-type new-FO reuse with frozen policies.

I. INTRODUCTION

Learning from demonstrations provides an effective way for robots to acquire complex manipulation skills. Recent visuomotor policies, such as Diffusion Policy (DP) and Action Chunking with Transformers (ACT), learn a mapping from visual observations and robot states to action sequences for executing manipulation tasks [1], [2]. These methods have achieved strong performance in various tasks by learning the correspondence between scene observations and demonstrated behaviors.

However, existing visuomotor policies mainly focus on learning *how to act*, while the identity of the manipulated object is usually implicitly encoded in the visual observation. Most existing settings assume that the task-relevant object is unique or visually distinct from other objects, allowing the policy to associate scene appearance with the corresponding action.

Fig. 1 compares RGB control, mask conditioning, and our method.

This assumption breaks down in realistic manipulation scenarios where multiple visually similar objects from the same category coexist. For example, several cans, packages, or industrial components may share similar appearance while representing different user-specified targets. In such cases, the fundamental challenge is no longer only whether the robot

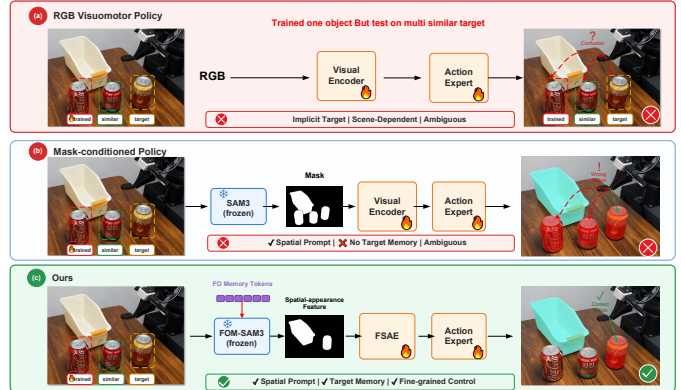


Fig. 1. **Fine-grained target selection.** (a) An RGB policy has no target input. (b) A mask gives only the target location. (c) FO Memory identifies the target, and FSAE conditions the policy.

can execute a manipulation skill, but whether it can explicitly identify *which object* the skill should be applied to. Existing policies lack an object identity interface, causing the learned skill to remain implicitly bound to the training-time target and making them prone to manipulating an incorrect Similar-FO during deployment.

Therefore, extending a learned manipulation skill to fine-grained objects requires more than improving action generation. It requires an explicit target identity interface that separates *which object to manipulate* from *how to manipulate it*. The key question is how to preserve an existing manipulation skill while enabling the robot to select and operate a user-specified object among visually similar candidates. Vision-Language-Action (VLA) models introduce language instructions as an additional interface for specifying manipulation goals. By combining visual observations with natural language, VLA models improve the flexibility of robot control and enable more general task specification.

However, language descriptions are often sparse compared with the visual details required for fine-grained object identification. The identity of a specific object may depend on local appearance cues, such as package patterns, textures, or instance-specific details, which are difficult to fully describe with a short language instruction. Therefore, language guidance alone may not provide a stable identity representation when multiple similar objects coexist.

This motivates a perception-control decoupled framework, where a vision foundation model provides explicit object-level understanding, while the action policy focuses on executing the

learned manipulation skill. Vision foundation models (VFM) provide a promising solution for the perception side of this problem. Large-scale pre-trained visual models contain rich semantic and appearance knowledge and have been widely used to improve robot perception. In particular, promptable segmentation models, such as the Segment Anything family, can provide explicit object localization through visual or language prompts [3], [4].

SAM3 further extends this capability toward concept-conditioned object localization. Given a prompt, SAM3 can produce structured outputs, including object masks, bounding boxes, and visual features [5]. These outputs provide a natural interface for robotic manipulation, since the action policy can focus on task-relevant object regions instead of the entire RGB observation.

However, a text prompt in a vision foundation model mainly represents a semantic concept rather than a persistent object identity. A short description can specify categories or attributes, but it cannot accurately preserve all visual evidence that distinguishes one physical instance from similar objects. For example, two cans from the same category may share similar shape and color, while their identity depends on subtle package patterns or local appearance details. Therefore, fine-grained manipulation requires a reusable visual identity memory beyond the original text prompt. To address this limitation, we propose **FO Memory Learning**, which converts a coarse concept prompt into a reusable fine-grained object memory while keeping SAM3 fully frozen. Instead of adapting the entire vision foundation model, we optimize only a small number of valid prompt tokens through a lightweight transformation in the original SAM3 prompt space.

The learned FO Memory captures instance-specific visual characteristics from a small set of multi-view registration images. After training, the optimized prompt states are exported as static FO Memory Tokens and stored in a memory bank. During deployment, an FO Prompt is used as an index to retrieve the corresponding memory, which directly conditions frozen SAM3 without registration images, online optimization, or model adaptation.

This design provides several advantages. First, it preserves the strong general visual knowledge of SAM3 by avoiding backbone modification. Second, adding a new FO only requires registering a new memory entry rather than collecting new action demonstrations or retraining the policy. Third, the same manipulation skill can be reused across different object appearances by explicitly switching the retrieved FO Memory Tokens. Although FO Memory enables explicit target selection, the action policy still requires a suitable visual condition for manipulation. Directly using full RGB observations may introduce irrelevant background information and visual distractors, requiring the policy to implicitly discover task-relevant regions.

We therefore introduce **Focused Spatial-Appearance Encoding (FSAE)**, which converts SAM3 outputs into object-centric visual conditions for downstream policies. FSAE uses SAM3 masks, boxes, and visual features to extract focused

representations of the manipulated object and its interaction context. Compared with full-image representation, this design provides the policy with explicit spatial and appearance information of task-relevant regions.

Furthermore, manipulation often depends on the relationship between the target object and its interaction region. For example, placing an object requires understanding both the target appearance and its relative position to the container. Within FSAE, **Target-Context Relational Encoding (TCRE)** models this interaction. TCRE combines target features, context features, and relative geometric information to produce relation-aware representations. The resulting features can be converted into compact visual conditions for Diffusion Policy or structured visual tokens for Action Chunking with Transformers, without changing their original action objectives. Together, FO Memory and FSAE establish a perception-control decoupled framework for fine-grained robotic manipulation. FO Memory provides persistent information about *which object should be manipulated*. Frozen SAM3 provides object localization and visual features describing *where the object is*. FSAE provides focused visual conditions that help the policy determine *how to act*.

This separation enables a learned manipulation skill to be reused across different fine-grained objects. A new FO can be incorporated by adding its FO Memory Tokens, while SAM3, FSAE, and the action policy remain unchanged. The framework therefore extends the target range of existing skills instead of relearning the manipulation behavior. We evaluate the proposed framework from both perception and manipulation perspectives. We first introduce FO30, a fine-grained object benchmark containing 30 objects from four coarse categories, and evaluate target localization, Similar-FO discrimination, and absent target rejection. We further conduct real-robot experiments with Diffusion Policy and ACT on manipulation tasks including Pick & Place, Push, and Assemble.

The experiments investigate whether FO Memory improves fine-grained identity selection, whether FSAE provides more effective visual conditions than full-image observations, and whether existing manipulation skills can be reused for newly registered objects without additional action demonstrations. Our contributions are summarized as follows:

- We propose **FO Memory Learning**, a lightweight fine-grained object memory mechanism built on frozen SAM3. By optimizing only valid prompt tokens with a residual and identity-initialized transformation, it converts coarse concept prompts into reusable FO Memory Tokens without vision foundation model adaptation.
- We introduce **Focused Spatial-Appearance Encoding (FSAE)** with Target-Context Relational Encoding (TCRE), which transforms SAM3 features, masks, and boxes into object-centric visual conditions. The proposed representation explicitly models target-context appearance and geometric relations and supports both compact DP conditions and ACT visual tokens.
- We establish a fine-grained perception and manipulation

evaluation framework with FO30 and real-robot experiments. The evaluation covers fine-grained object identification, Similar-FO discrimination, absent-target rejection, visual distractor robustness, and new-FO skill reuse with frozen policies.

Sec. III presents the method, and Sec. IV presents the experiments.

II. RELATED WORK

A. Object-Centric Visuomotor Policy Learning

Recent visuomotor policies have achieved strong manipulation performance by learning a direct mapping from visual observations to robot actions. Diffusion Policy (DP) formulates visuomotor control as conditional action generation with diffusion models, enabling multimodal trajectory modeling and stable learning for complex manipulation tasks [1]. Action Chunking with Transformers (ACT) further improves imitation learning by predicting action sequences through transformer-based sequence modeling [2]. More recently, vision-language-action (VLA) models incorporate large-scale visual and language knowledge to provide flexible task specification [6]–[8].

Despite their success, most visuomotor policies still rely on global image representations as visual conditions. The identity of the manipulated object is therefore implicitly entangled with scene appearance, object location, and background context. This design is effective when the target object is unique, but becomes unreliable when multiple visually similar objects co-exist. In such cases, a policy may generate a valid manipulation trajectory while selecting the wrong object instance.

To reduce irrelevant visual information, recent works explore object-centric representations for robot manipulation. VIOLA introduces object-centric representations from visual demonstrations to improve generalization across manipulation tasks [9]. GROOT learns object-centric 3D representations for robot manipulation and demonstrates improved robustness to environmental variations [10]. RoboGround uses visual grounding information to provide task-relevant object regions for manipulation policies [11]. Other approaches combine segmentation or semantic representations with diffusion policies to improve generalization under cluttered scenes and object variations [12]–[14].

However, these approaches mainly focus on extracting *where* the object is or reducing background interference. They do not explicitly maintain a persistent identity representation for distinguishing one user-specified instance from visually similar objects. In contrast, our work introduces FO Memory to provide an explicit identity interface and uses FSAE to transform object-level perception into policy-compatible visual conditions.

B. Promptable Vision Foundation Models and Fine-Grained Object Memory

Vision foundation models have significantly advanced open-world visual understanding. Segment Anything Model (SAM) introduces a promptable segmentation framework that enables flexible object localization with points, boxes, and masks [3].

SAM2 extends this capability to video scenarios by introducing a unified image-video segmentation framework with memory-based tracking [4]. More recently, SAM3 further improves concept-conditioned segmentation by incorporating text-driven object localization and richer visual representations [5].

These models provide an effective perception interface for robotics because they can generate explicit object masks, bounding boxes, and visual features. Several works have explored integrating SAM-based representations into robot learning. SAM-E connects SAM features with sequential imitation learning by extracting object-aware visual representations [12]. Other approaches combine visual grounding, segmentation, and policy learning to improve manipulation robustness under object and scene variations [11].

However, generic prompts in vision foundation models mainly describe semantic concepts rather than persistent object identities. A text description such as “a red can” or “a cup” may locate objects belonging to a category, but cannot reliably preserve the subtle appearance differences between individual instances. This limitation becomes critical in robotic manipulation, where the robot must execute an existing skill on a specific user-selected object among multiple similar candidates.

Recent prompt learning and representation adaptation methods attempt to improve model specialization by optimizing prompt tokens or visual embeddings [15]–[17]. Nevertheless, these methods mainly adapt the perception model for recognition or segmentation and do not provide a reusable identity memory that can directly interface with downstream robot policies.

In contrast, our FO Memory Learning stores fine-grained object identity in the original SAM3 prompt space while keeping the vision foundation model fully frozen. The learned memory can be retrieved by an FO identifier and reused across episodes, enabling a frozen manipulation policy to explicitly select and operate on the requested object instance. The overall framework is shown in Fig. 2.

III. METHOD

Sec. III-A defines the inputs and execution order. Secs. III-B–III-C describe FO Memory Learning and localization. Secs. III-D, III-E, and III-F describe FSAE and its policy interfaces. Sec. III-G gives the closed-loop procedure. Implementation details are in Secs. D and E.

A. Problem Definition and Execution Order

At control step t , the system receives synchronized RGB images $I_t^c \in \mathbb{R}^{H \times W \times 3}$ from $N_c = 2$ wrist cameras, robot state $r_t \in \mathbb{R}^{D_r}$, and two ordered FO identifiers. Slot $k = 0$ is the manipulation target and slot $k = 1$ is the task context or receptacle. The policy predicts

$$\hat{A}_t = \pi_\theta(O_t) \in \mathbb{R}^{T_a \times D_a}, \quad D_r = D_a = 10, \quad (1)$$

where the observation history and T_a depend on the DP or ACT backbone. Computation proceeds as FO registration and memory export, memory-conditioned localization, cache

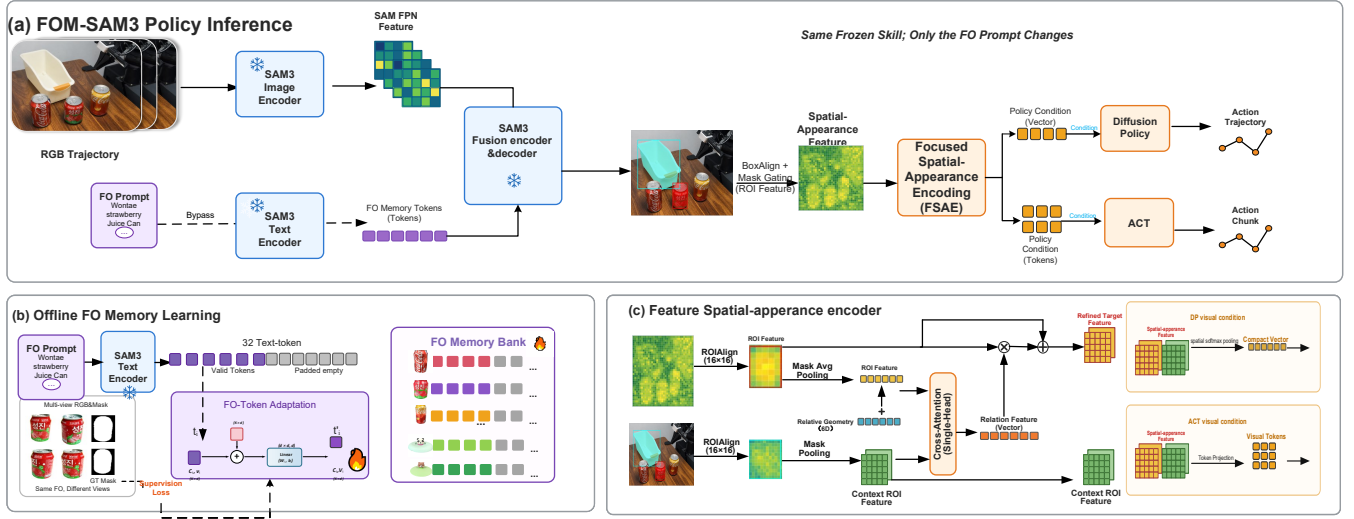


Fig. 2. **Overview of FOM-SAM3 and FSAE.** (a) An FO Prompt retrieves FO Memory Tokens. Frozen SAM3 localizes the target. FSAE conditions DP or ACT. (b) Offline FO Memory Learning updates only valid prompt tokens. SAM3 remains frozen. (c) FSAE uses ROI extraction, mask gating, task context, and relative geometry. It outputs a compact vector for DP or visual tokens for ACT.

construction, FSAE encoding, action prediction, and closed-loop tracking. Eq. (1) defines the common policy output.

This design separates VFM perception from policy control. FOM-SAM3 specifies and localizes the queried FO. The action policy predicts the manipulation from the resulting object condition.

B. FO Registration and Memory Export

For FO i , the registration set is

$$\mathcal{R}_i = \{(I_i^{(n)}, Y_i^{(n)})\}_{n=1}^{N_i}. \quad (2)$$

Eq. (2) collects the registration views and masks for one FO.

The frozen SAM3 text encoder is run once on user-assigned FO Prompt p_i . It returns $C_i \in \mathbb{R}^{32 \times 256}$ and its native padding mask. The valid positions are

$$(C_i, m_i^{\text{text}}) = \mathcal{E}_{\text{text}}^{\text{frozen}}(p_i), \quad V_i = \{j \mid m_{i,j}^{\text{text}} = 0\}. \quad (3)$$

Eq. (3) selects the non-padding prompt slots. Only these valid positions are optimized:

$$\tilde{C}_{i,V_i} = W_i(C_{i,V_i} + \Delta_i) + b_i, \quad \tilde{C}_{i,\bar{V}_i} = C_{i,\bar{V}_i}. \quad (4)$$

In Eq. (4), Δ_i, b_i are zero-initialized and W_i is identity-initialized. Padding positions and the native geometry prompt stay frozen. For 4–8 valid tokens, one FO has 66,816–67,840 trainable parameters; all SAM3 parameters remain frozen.

FO Memory Learning operates only in the SAM3 prompt space. It does not finetune SAM3 or relearn its visual encoder.

Training scenes are generated from registration RGB/mask pairs. A positive scene contains the queried FO and a configured same-group FO. A hard negative removes the query but retains that same-group FO. Positive candidates are assigned by Hungarian matching with object, box-L1, and GIoU costs 2, 5, 2. The optimized loss is

$$\mathcal{L}_{\text{mem}}^i = \mathcal{L}_+ + 0.1\mathcal{L}_- - \mathcal{L}_{\text{pres}}^-, \quad (5)$$

$$\begin{aligned} \mathcal{L}_+ = & \mathcal{L}_{\text{obj}} + 0.8\mathcal{L}_{\text{Dice}} + 0.2\mathcal{L}_{\text{BCE}} \\ & + 0.1\mathcal{L}_{\text{L1}} + 0.1\mathcal{L}_{\text{GIoU}} + \mathcal{L}_{\text{pres}}^+. \end{aligned} \quad (6)$$

In Eq. (6), a hard-negative scene supervises only the native presence pathway; the visible same-group FO is not assigned as a negative mask target. The learned valid slots form a persistent FO identity representation. The representation is reused across scenes and episodes. After checkpoint selection, Eq. (4) is evaluated once. The static artifact stores final valid-token values and positions, text mask, frozen geometry state, memory ID, and provenance hashes. Deployment performs only

$$M_{\text{FO}}^i = \mathcal{B}_{\text{FO}}[\text{id}(p_i)], \quad (7)$$

The lookup in Eq. (7) runs without the text encoder or Δ_i, W_i, b_i .

C. Memory-Conditioned Localization and Cache

The retrieved memory directly conditions frozen SAM3. Its SAM feature, mask, and box are frozen pre-trained visual observations. They are not relearned by the action policy. For each image it returns 200 candidates, native presence logit s_t^{pres} , and FPN feature H_t^c . Candidate q is ranked by

$$s_{t,q}^{\text{joint}} = \sigma(s_t^{\text{pres}}) \sigma(s_{t,q}^{\text{obj}}). \quad (8)$$

The score in Eq. (8) uses joint threshold 0.5, mask threshold 0.5, and mask-NMS IoU 0.8. The accepted candidate supplies the mask and normalized box.

Policy demonstrations are processed with the frozen 336-pixel backbone and corrected bidirectional tracks. For each camera and frame, the cache is

$$C_t^c = \left(H_t^c, \{Y_t^{c,k}, B_t^{c,k}, v_t^{c,k}\}_{k=0}^1 \right), \quad (9)$$

In Eq. (9), $H_t^c \in \mathbb{R}^{256 \times 24 \times 24}$ is backbone_fpn.2, $Y_t^{c,k} \in [0, 1]^{24 \times 24}$, $B_t^{c,k} = [c_x, c_y, w, h] \in [0, 1]^4$, and $v_t^{c,k}$ is slot validity. Missing objects keep their ordered slot and use zero mask/box. The wrapper broadcasts each box as constant planes, denoted by $\Pi(B)$, and packs

$$X_t^c = [H_t^c, Y_t^{c,0}, Y_t^{c,1}, \Pi(B_t^{c,0}), \Pi(B_t^{c,1})] \in \mathbb{R}^{266 \times 24 \times 24}. \quad (10)$$

The packed tensor in Eq. (10) is cached in bfloat16 and converted to policy compute dtype on loading. SAM3 is outside the policy graph; only FSAE and the action model receive action-loss gradients.

D. Focused Spatial-Appearance Encoding

FSAE is a trainable perception-to-control adapter. It converts frozen SAM3 outputs into policy-compatible object conditions. FSAE and the action policy are optimized jointly by the imitation loss.

Each normalized box is converted to $x_1 y_1 x_2 y_2$ on the 24×24 grid. ROIALign extracts a feature and aligned hard mask,

$$R_t^{c,k} = \text{ROIALign}(H_t^c, B_t^{c,k}) \in \mathbb{R}^{256 \times 16 \times 16}, \quad (11)$$

$$U_t^{c,k} = \mathbb{I}[\text{ROIALign}(Y_t^{c,k}, B_t^{c,k}) \geq 0.3]. \quad (12)$$

FSAE uses mask-normalized pooling

$$\text{MP}(R, U; G) = \frac{\text{AvgPool}_G(R \odot U)}{\text{AvgPool}_G(U) + \epsilon}. \quad (13)$$

Eq. (12) defines ROI extraction and mask alignment. Eq. (13) removes the effect of empty mask area.

The two ordered slots retain the target and its task-related context. The target provides one normalized 256-D query summary. The context is pooled to a 4×4 grid, flattened to 16 normalized tokens, and used as keys/values. Relative geometry is

$$g = \left[\frac{c_x^0 - c_x^1}{w^1}, \frac{c_y^0 - c_y^1}{h^1}, \log \frac{w^0}{w^1}, \log \frac{h^0}{h^1}, \text{IoU}(B^0, B^1), \rho_{0 \in 1} \right], \quad (14)$$

In Eq. (14), the first four entries are clipped to $[-4, 4]$, and $\rho_{0 \in 1}$ is target-mask support inside the context box. With relation width $d = 32$,

$$q = W_q \text{LN}(\text{MP}(R^0, U^0; 1)) + W_g g, \quad (15)$$

$$[k_j, v_j] = W_{kv} \text{LN}(h_{1,j}), \quad \alpha_j = \text{softmax}_j(q^\top k_j / \sqrt{d}), \quad (16)$$

$$r = \sum_{j=1}^{16} \alpha_j v_j. \quad (17)$$

In Eq. (17), invalid context cells are masked before softmax. The target-only update is

$$\tilde{R}_t^{c,0} = R_t^{c,0} + R_t^{c,0} \odot \tanh(W_r r + b_r) \odot U_t^{c,0} v_t^{c,0} v_t^{c,1}, \quad (18)$$

$$\tilde{R}_t^{c,1} = R_t^{c,1}. \quad (19)$$

In Eq. (19), W_r, b_r are zero-initialized. Missing either slot therefore gives an exact identity map. The shared relation refinement has 33,344 trainable parameters. This residual update preserves the SAM3 target feature while adding task-specific cues learned from demonstrations.

E. Diffusion Policy Interface

The same refined features support two policy interfaces. DP uses a compact vector, while ACT retains local visual tokens.

A 1×1 convolution maps each refined ROI to 16 heatmaps. Mask-aware SpatialSoftmax returns 32 expected coordinates per object. We use $[\cdot]_{c,k}$ to denote concatenation over cameras and object slots:

$$z_t^{\text{DP}} = [\text{SSM}(\tilde{R}_t^{c,k}, U_t^{c,k}), B_t^{c,k}]_{c=1,2; k=0,1} \in \mathbb{R}^{144}. \quad (20)$$

The vector in Eq. (20) and states r_{t-1}, r_t form a 164-D condition. The 24-step training window is $A_t = [a_{t-1}, \dots, a_{t+22}]$. DP minimizes

$$A_t^\tau = \sqrt{\bar{\alpha}_\tau} A_t + \sqrt{1 - \bar{\alpha}_\tau} \epsilon, \quad (21)$$

$$\mathcal{L}_{\text{DP}} = \mathbb{E} \|\epsilon - \epsilon_\theta(A_t^\tau, \tau, [z_t^{\text{DP}}, r_{t-1}, r_t])\|_2^2. \quad (22)$$

Eq. (22) is the DP training objective. At inference, positions 1:17 correspond to actions $a_t : a_{t+15}$, followed by replanning.

F. ACT Interface

Each refined ROI is mask-normalized to 8×8 . Its 256-D cells are projected to 512-D and receive local 2-D position, box, camera, and role embeddings. Thus,

$$Z_t^{\text{ACT}} \in \mathbb{R}^{256 \times 512} \quad (2 \text{ cameras} \times 2 \text{ slots} \times 64 \text{ cells}). \quad (23)$$

Eq. (23) defines the ACT visual-token condition. Adding latent and current-state tokens gives 258 ACT encoder tokens. The training VAE encodes CLS, current state, and 30 actions into $(\mu_t, \log \sigma_t^2) \in \mathbb{R}^{32}$. For padding mask p_j ,

$$\mathcal{L}_{\text{ACT}} = \frac{\sum_j (1 - p_j) \|a_{t+j} - \hat{a}_{t+j}\|_1}{\sum_j (1 - p_j)} + 10 D_{\text{KL}}(q(z_t | r_t, A_t) \| \mathcal{N}(0, I)). \quad (24)$$

Eq. (24) is the ACT training objective. Inference uses the standard zero prior latent and temporal ensembling and emits one action per control step.

G. Closed-Loop Deployment

At reset, target and context IDs retrieve their memories. The frozen 504-pixel detector initializes both slots in each wrist view, followed by one 336-pixel tracker per camera. Each control step propagates the ordered masks/boxes, computes the frozen FPN feature, runs FSAE and DP/ACT, and executes in receding-horizon mode at 5 Hz. Persistent FO memory and episode-local tracker memory are separate. If a required slot becomes invalid, actuation is suspended and the same FO memories reinitialize the trackers before execution resumes. All deployed models remain frozen. The FO memory bank stores long-term identity across episodes. The tracker memory bank maintains short-term continuity only within one episode.

IV. EXPERIMENTS

Sec. IV-A gives the common setup. Secs. IV-B and IV-C evaluate perception and robot manipulation. Secs. IV-D and IV-E report the ablation and generalization tests. Sec. IV-F summarizes failure cases and limitations.

A. Setup and Protocol

1) Platform: We use a single-arm AgileX PiPER robot equipped with two Intel RealSense D405 RGB cameras. One camera is mounted on the wrist for the manipulation view, while the other provides a fixed side view for complementary visual observations. We collect 120 human-teleoperated demonstrations for each task. The policies use two wrist RGB views and the robot state.

2) Baselines: We choose the 2D image-based Diffusion Policy (RGB-DP) [1] and ACT (RGB-ACT) [2] as standard visuomotor baselines. We also compare S²-Diffusion [13], we follow its original implementation and adapt it to use two-camera inputs with mask and depth observations. Ours-DP and Ours-ACT use the same policy architectures, training data, and optimization settings as their corresponding baselines. The only difference is the visual condition. All policies are trained from scratch, and no model parameters are updated during evaluation. Detailed implementation and training configurations are provided in the Appendix.

3) Tasks and Protocol: We design three manipulation tasks: Can Pick & Place, Box Push, and Lid Assembly to study fine-grained target selection when multiple visually similar objects coexist. During training, each task contains only one unique target object and its corresponding task context, allowing the policy to learn the manipulation skill without object ambiguity. During evaluation, we introduce Similar-FOs and visual distractors while keeping the learned policy unchanged. A trial succeeds only when the robot selects the queried FO and completes the full manipulation task. Wrong-FO selection, incorrect interaction, timeout, or incomplete execution are counted as failures. Representative task sequences and evaluation objects are shown in Fig. 3.

B. Fine-Grained FO Perception

1) Dataset: To study fine-grained FO perception and memory, we build FO30 with 30 FOs from four coarse groups. For

TABLE I
FO PERCEPTION RESULTS. WE COMPARE SAM3 TEXT PROMPT, TEXT-TOKEN OPTIMIZATION, PTSAM, M2C, AND FO MEMORY LEARNING UNDER THE SAME FROZEN-SAM3 PROTOCOL. FO MEMORY LEARNING OUTPERFORMS ALL ALTERNATIVES ON BOTH FO30 TEST SETS.

Method	cgF1↑	pmF1↑	IL-MCC↑	Joint FPR↓
<i>FO30-S: Standard</i>				
SAM3 Text Prompt [5]	24.30	73.57	0.3304	18.96
Text-Token Opt. [15]	35.87	81.49	0.4403	17.19
PTSAM [16]	26.30	73.37	0.3585	15.75
M2C [17]	35.82	81.53	0.4394	17.26
FO Memory Learning (Ours)	62.38	93.56	0.6668	6.85
<i>FO30-C: Challenge</i>				
SAM3 Text Prompt [5]	20.34	55.89	0.3641	14.42
Text-Token Opt. [15]	29.41	62.76	0.4687	13.06
PTSAM [16]	20.82	56.00	0.3718	12.63
M2C [17]	29.77	62.95	0.4729	13.10
FO Memory Learning (Ours)	55.98	85.42	0.6554	7.60

each FO, we capture 60 single-object registration images from different views; FO30-S contains 600 standard test images, FO30-C contains 400 harder images with more Similar-FO co-occurrence, and full construction details are given in Sec. B.

2) Baselines and Metrics: We compare SAM3 Text Prompt, Text-Token Optimization, PTSAM [16], M2C [17], and our FO Memory Learning. We reproduce PTSAM and M2C in the same frozen-SAM3 training and evaluation framework. All methods use the same frozen SAM3, test data, thresholds, NMS, and evaluator. Each test image is queried with all 30 FO Prompts without a GT mask, point, or box. For a direct comparison, we use the official SAM3 CGF1 evaluator and report pmF1, IL-MCC, and cgF1 [5]. pmF1 measures positive-mask quality, while IL-MCC measures image-level present/absent decisions. Their combined score is

$$\text{cgF1} = \text{pmF1} \times \text{IL-MCC}. \quad (25)$$

Joint FPR is an additional diagnostic for false acceptance when the queried FO is absent. Detailed metric definitions are provided in Sec. G-B. Additional FO30 results are provided in Secs. G and G-A.

Table I compares the five prompt routes on both FO30 test sets.

3) Results: FO Memory Learning gives the best result on both test sets. On FO30-S, it improves pmF1 by 12.03 points and cgF1 by 26.51 points over the strongest completed alternative. It also lowers Joint FPR from 15.75% to 6.85%. On FO30-C, it improves pmF1 by 22.47 points and cgF1 by 26.21 points. Its Joint FPR is 7.60%, compared with 12.63% for the lowest alternative. The harder split still reduces our pmF1 from 93.56% to 85.42%. Similar-FO FPR also increases from 13.77% to 18.57%, as detailed in Table XIX.

4) Prompt-Route Runtime: We compare the two prompt routes with some 504-pixel images and the same frozen SAM3. After five warm-up passes, 30 measurements give 48.18 ± 4.11 ms for the SAM3 Text Prompt and 43.26 ± 5.17 ms for preloaded FO Memory Tokens. The memory route skips online text encoding and reduces the mean localization latency by 4.92 ms.



Fig. 3. **Real-robot tasks and evaluation objects.** The rows show Pick & Place, Push, and Assemble. The right column shows the trained FO, a Similar-FO, and task distractors.

TABLE II

ABSENT-FO REJECTION IN REAL-ROBOT SCENARIOS. THE QUERIED FO IS ABSENT WHILE SAME-CATEGORY SIMILAR-FOS REMAIN. A SUCCESSFUL TRIAL REQUIRES REJECTING THE QUERY AND BLOCKING POLICY EXECUTION. RESULTS ARE AVERAGED OVER THREE MANIPULATION TASKS.

Method	Single Similar-FO	Multiple Similar-FOs
SAM3 Text Prompt	2/10	0/10
FOM-SAM3	10/10	8/10

5) FO Rejection Evaluation: Open-vocabulary perception often selects the candidate with the highest semantic similarity. If the queried FO is absent but a Similar-FO remains, this behavior can produce a false positive and trigger an action on the wrong object. A robot must determine both where the target is and whether it is present.

We evaluate whether the system can reject an absent queried FO instead of forcing a match to a visually similar object. For each manipulation task, the queried FO is removed from the scene while one or multiple same-category Similar-FOs remain. The system is expected to output no valid target and prevent the action policy from executing.

We consider two failure cases. First, a *false positive* occurs when the perception module incorrectly localizes a visible Similar-FO as the queried FO. Second, an execution failure occurs when a wrong FO is detected and the policy continues to execute the manipulation. A trial is successful only when the system correctly rejects the absent FO and blocks further action execution. For successful manipulation trials, the robot must also select the queried FO and complete the entire task execution.

We evaluate two settings: *Single Similar-FO*, where one same-category Similar-FO remains, and *Multiple Similar-FOs*, where multiple candidates coexist. Results are averaged over the three manipulation tasks. Table II reports the absent-query

rejection accuracy.

The text-prompt baseline often matches the most similar visible candidate. FO Memory represents the fine-grained identity rather than only category semantics. When the queried FO is absent, its FO Memory Tokens do not match the remaining Similar-FOs. FOM-SAM3 suppresses these candidates and rejects the query instead of forcing a match. This persistent identity constraint prevents the policy from acting on the wrong object.

C. Real-Robot Manipulation

1) Test Conditions: All policies are trained with a single seen target object for each task. Figure 3 shows the three tasks, seen targets, Similar-FOs, and distractors. All conditions use the same five target markers and ten rollouts per method-task pair.

ID uses the seen target, training background, and task context without extra distractors. The target is moved across the five fixed markers. This setting tests whether each policy learned the basic manipulation skill.

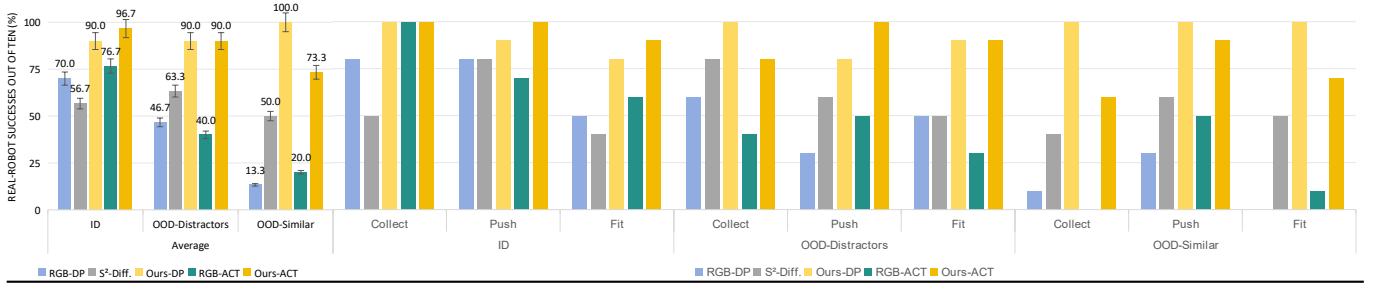
OOD-Distractors changes the background and adds an unrelated object from a different coarse category. The first five rollouts use distractor 1, and the last five use distractor 2. The distractor stays fixed while the seen target moves across the five markers.

OOD-Similar changes the background and places a same-category Similar-FO beside the queried target. The first five rollouts use Similar-FO 1, and the last five use Similar-FO 2. The Similar-FO stays fixed while the queried target moves across the same markers. The robot must select the queried FO and complete the task. A correct action on the wrong object is a failure.

Table III compares all methods under the three test conditions.

TABLE III

REAL-ROBOT MANIPULATION RESULTS. WE COMPARE FIVE METHODS ON THREE TASKS UNDER ID, OOD-DISTRACTORS, AND OOD-SIMILAR CONDITIONS. OURS-DP AND OURS-ACT GIVE THE STRONGEST OVERALL ROBUSTNESS, ESPECIALLY UNDER OOD-SIMILAR.



Method	ID			OOD-Distractors			OOD-Similar		
	Pick&Place	Push	Assemble	Pick&Place	Push	Assemble	Pick&Place	Push	Assemble
RGB-DP [1]	8/10	7/10	5/10	6/10	3/10	5/10	1/10	3/10	0/10
S²-Diff. [13]	10/10	7/10	4/10	8/10	6/10	5/10	4/10	6/10	5/10
Ours-DP	10/10	9/10	8/10	10/10	8/10	9/10	10/10	10/10	10/10
RGB-ACT [2]	10/10	8/10	6/10	4/10	5/10	3/10	0/10	5/10	1/10
Ours-ACT	10/10	9/10	9/10	8/10	10/10	9/10	8/10	9/10	7/10

2) Results: All methods achieve good performance under ID, showing that they learn the required manipulation skills from demonstrations.

Under OOD-Distractors, RGB-based policies are affected by additional irrelevant objects in the scene. Our method improves robustness by providing object-focused visual conditions through FSAE instead of relying only on the global image representation.

OOD-Similar is the main setting of this work. When multiple Similar-FOs coexist with the target object, standard visuomotor policies lack an explicit fine-grained object identity representation. Although they can generate valid manipulation trajectories, they may confuse the target with a Similar-FO and execute the action on the wrong object.

FO Memory provides explicit target identity, while FSAE provides focused visual conditions for action generation. Both DP and ACT benefit from this design and achieve better performance under Similar-FO interference.

These results show that our framework separates object identification from skill execution. It enables existing visuomotor policies to reuse learned manipulation skills while operating on the specified fine-grained object.

Insight: A policy trained with one target has no explicit identity constraint when several Similar-FOs appear at test time. Fine-grained object identity should be grounded before action generation. FO Memory determines *which object*, while FSAE represents *how to act on it*.

D. Ablation of Visual Condition Representation

1) Compared Conditions: We evaluate different visual condition designs under the OOD-Distractors setting, where the target object remains identifiable but irrelevant background objects are introduced. To isolate the effect of the policy input representation, all variants use the same SAM3 Text Prompt for target localization, the same Diffusion Policy backbone, and identical demonstrations. Only the visual condition provided

TABLE IV

VISUAL-CONDITION ABLATION. ALL VARIANTS USE THE SAME SAM3 TEXT PROMPT, DIFFUSION POLICY, DEMONSTRATIONS, AND OOD-DISTRACTORS SETTING. FSAE GIVES THE BEST RESULT ACROSS ALL THREE TASKS.

Visual Condition	Pick&Place	Push	Assemble
RGB Feature (ResNet)	6/10	3/10	5/10
Mask-guided RGB (ResNet)	8/10	6/10	5/10
SAM3 Object Feature w/o TCRC	10/10	7/10	8/10
FSAE (Ours)	10/10	8/10	9/10

to the policy is changed. FO Memory is not evaluated in this ablation. RGB Feature directly encodes the full image and allows the policy to infer task-relevant regions implicitly. Mask-guided RGB Feature uses the SAM3 mask to suppress background regions. SAM3 Object Feature extracts frozen SAM3 features within the localized object region. FSAE further incorporates focused target appearance, manipulation context, and relative geometry for action prediction.

Table IV isolates the effect of the visual condition.

2) Results: The results show that progressively introducing object-centric visual conditions improves manipulation robustness under background disturbances. RGB Feature relies on global scene representations, requiring the policy to implicitly identify task-relevant regions and making it vulnerable to irrelevant visual cues. Mask-guided RGB reduces background interference by explicitly highlighting the target region. SAM3 Object Feature further improves performance by providing pretrained object-level representations with stronger spatial focus. Compared with object features alone, FSAE achieves the best results by further modeling target appearance, manipulation context, and their relative geometry, providing a more informative visual condition for action prediction.

E. Generalization

Mask-DP augments RGB-DP with SAM3 text-prompted masks. Ours-DP combines FOM-SAM3 and FSAE to provide identity-aware and object-centric visual conditions.

TABLE V

NEW-FO REUSE WITH FROZEN POLICIES. WE CHANGE ONLY THE RETRIEVED FO MEMORY TOKENS AND KEEP FOM-SAM3, FSAE, AND EACH POLICY FROZEN. BOTH POLICY BACKBONES REUSE THE LEARNED SKILLS ON NEWLY REGISTERED SIMILAR-FOs.

Method	Pick&Place	Push	Assemble
Ours-DP	10/10	10/10	8/10
Ours-ACT	7/10	8/10	8/10

TABLE VI

SIMILAR-FO AND INSTANCE-VARIATION GENERALIZATION. WE COMPARE RGB-DP, MASK-DP, AND OURS-DP WITH ONE FROZEN-POLICY ROLLOUT PER OBJECT. OURS-DP SUCCEEDS ON ALL TESTED SIMILAR-FOs AND NEW INSTANCES.



Method	Similar-FOs			Novel Instances		
	FO-1	FO-2	FO-3	Obj-A	Obj-B	Obj-C
RGB-DP	×	×	×	×	×	×
Mask-DP	×	✓	×	✓	×	×
Ours-DP	✓	✓	✓	✓	✓	✓

1) New FO Reuse with Frozen Policy: We evaluate whether FO Memory enables frozen policies to manipulate newly registered Similar-FOs without additional demonstrations or policy updates. For each newly registered FO, we place it at five different spatial configurations and record the success rate over trials. During evaluation, the policy remains completely frozen, and only the retrieved FO Memory Tokens are replaced according to the queried object. Table V reports this within-type reuse test. The results show that FO Memory enables frozen manipulation policies to directly transfer to newly registered Similar-FOs. Unlike conventional end-to-end DP or ACT policies, which implicitly associate visual observations with actions and lack an explicit mechanism to specify which object instance should be manipulated, our approach separates object identity grounding from policy execution. By only replacing the retrieved FO Memory Tokens, the same policy can successfully interact with unseen target objects without collecting new demonstrations or retraining. This verifies that FO Memory expands the reachable target set while preserving the learned interaction skills.

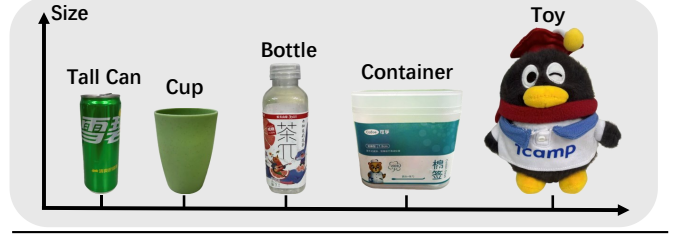
2) Similar-FO and Instance Variation Generalization: We evaluate instance-level manipulation generalization with a frozen policy. The Similar-FO setting tests whether the policy can select the queried target among same-category objects with similar appearance. The Novel Instances setting tests whether the same policy can handle different object appearances while preserving the learned manipulation skill. Each object is evaluated with one independent rollout.

Table VI compares target selection and appearance variation.

RGB-DP does not preserve the queried identity when Similar-FOs appear together. Mask-DP provides explicit localization but still lacks instance-level identity. Ours-DP retains

TABLE VII

INSTANCE GENERALIZATION WITH A FROZEN POLICY. WE EVALUATE FIVE PICK & PLACE OBJECTS WITH DIFFERENT APPEARANCE AND SCALE, USING ONE ROLLOUT PER OBJECT. OURS-DP SUCCEEDS ON ALL FIVE TESTED OBJECTS.



Method	Obj-A	Obj-B	Obj-C	Obj-D	Obj-E
RGB-DP	✓	✓	×	×	✓
Mask-DP	✓	✓	✓	×	×
Ours-DP	✓	✓	✓	✓	✓

TABLE VIII

SEQUENTIAL TARGET MANIPULATION WITH SIMILAR-FOs. WE COMPARE RGB-DP, MASK-DP, AND OURS-DP WHILE THE QUERIED TARGET CHANGES ACROSS THE THREE TASKS. ONLY OURS-DP COMPLETES THE FULL SEQUENCE WITH CORRECT TARGET SELECTION.

Method	Pick&Place	Push	Assemble
RGB-DP	×	×	×
Mask-DP	×	✓	×
Ours-DP	✓	✓	✓

the queried FO identity and also handles the tested appearance variations. This is instance-level manipulation generalization with a frozen policy, not open-world skill learning.

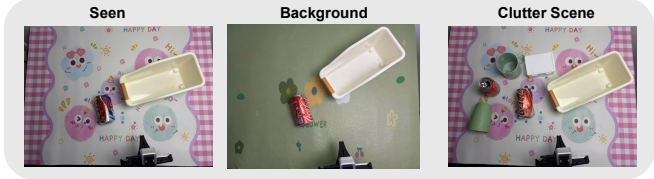
3) Instance Variation Generalization: We use Pick & Place to test larger changes in object size, shape, and appearance. The manipulation skill and policy remain fixed. Each everyday object is registered as a new FO and evaluated with one independent rollout. Success requires grasping the queried object and completing the placement.

Table VII reports the five object trials.

This experiment tests whether object-centric conditions remain useful under larger appearance and scale variations beyond Similar-FO discrimination.

Sequential Target Manipulation with Multiple Similar-FOs: We evaluate whether the proposed method can maintain target identity during long-horizon manipulation with multiple Similar-FOs. Multiple visually similar objects are placed in the same scene, and the robot is required to manipulate different target objects following a predefined instruction sequence. At each step, the queried FO is changed according to the sequence, requiring the system to re-identify the specified object before executing the learned skill. A trial is considered successful only when all manipulation steps are completed with correct target selection and execution. Any incorrect object selection at any step causes the entire sequence to fail. Table VIII reports the complete sequence result. The results show that FO Memory enables reliable long-horizon manipulation when multiple Similar-FOs coexist in the same scene. During sequential execution, the robot must repeatedly identify the specified target rather than relying on a fixed visual-action

TABLE IX
BACKGROUND GENERALIZATION ON PICK & PLACE. WE COMPARE THREE METHODS ON ONE SEEN AND TWO UNSEEN VISUAL SETTINGS WITH A FIXED TARGET AND POLICY. OURS-DP GIVES THE STRONGEST ROBUSTNESS TO BACKGROUND AND CLUTTER CHANGES.



Method	Seen	Background	Clutter
RGB-DP	8/10	5/10	0/10
Mask-DP	10/10	10/10	2/10
Ours-DP	10/10	10/10	9/10

association. RGB-DP lacks explicit object identity grounding and can easily select an incorrect Similar-FO after the target changes. Mask-DP provides spatial localization through object masks but cannot preserve fine-grained identity among visually similar objects. In contrast, our method retrieves target-specific FO Memory and provides consistent object-centric conditions to the policy, allowing the frozen policy to complete the entire manipulation sequence with correct target selection.

Background Generalization: We evaluate robustness to background variations using the Pick & Place task. The target object, manipulation skill, and policy remain unchanged, while only the background is modified. The evaluation includes one seen background and two unseen backgrounds. Each setting is tested with 10 trials, and success requires correctly grasping and placing the target object. The visual settings and success rates are shown in Table IX.

Our method achieves consistent performance across different backgrounds, demonstrating improved robustness to environmental appearance changes. RGB-DP relies on global visual correlations and is sensitive to background variations. Mask-DP reduces background interference through explicit object localization but lacks fine-grained object-aware representation. By leveraging FO-specific representations, our method focuses on the target object and maintains stable manipulation performance under unseen backgrounds.

F. Failure Cases and Limitations

We group the observed failures into four cases.

- 1) Tracking Failure:** The initial localization is correct, but the tracker later loses the target. This occurs under severe occlusion, large viewpoint change, or fast motion. The policy then receives an incomplete or outdated object condition.
- 2) Perception Failure:** FOM-SAM3 may miss the queried FO or select a Similar-FO. This occurs for highly ambiguous objects or under weak registration coverage.
- 3) Manipulation Failure:** The queried FO is detected and tracked, but the policy does not finish the task. Typical causes include grasp error, contact dynamics, poor alignment, and execution drift.
- 4) Generalization Boundary:** New-FO reuse requires compatible appearance and interaction geometry. Large appearance

shifts, new interaction types, or unseen viewpoints may require more registration views. The policy and SAM3 still remain frozen.

Perception failures include misses and false acceptance of a Similar-FO. Robot failures can occur at localization, tracking, target selection, grasp/contact, transport/alignment, or timeout. The finalized aggregate record does not preserve an audited failure-stage label for every rollout, so no numerical causal decomposition is claimed. Ours-ACT fails 4/10 OOD-Similar Pick & Place trials, and both variants fail 2/10 within-type Assemble trials; these failures are retained in the reported results.

Completed robot cells contain ten rollouts for one trained configuration, not independent policy-training seeds. Exact task tolerances and timeout values were not retained in the final summary and are not reconstructed. Current trajectory JSONL files lack immutable per-rollout outcome labels sufficient to regenerate every aggregate. No matched ablation isolates mask gating, box geometry, or relation refinement. Finally, new-FO reuse is bounded by compatible interaction geometry and does not constitute acquisition of a new skill.

V. CONCLUSION AND LIMITATIONS

This work introduces FOM-SAM3 for persistent fine-grained target memory and FSAE for constructing ordered target/context conditions for DP and ACT. FO30 results show improved target segmentation, identity ranking, and absent-query rejection with frozen SAM3. Completed real-robot evaluations show stronger robustness to ordinary distractors and Similar-FOs, and within-type tests demonstrate query switching among newly registered FOs without policy retraining. The evidence remains limited by ten-rollout aggregates, absent cross-seed policy statistics, and the lack of matched FSAE component ablations; the reported reuse result therefore applies only to the tested within-type, geometry-compatible objects.

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APPENDIX A HARDWARE AND RUNTIME SETUP

Table X summarizes the deployment setup.

TABLE X
REAL-ROBOT HARDWARE AND RUNTIME CONFIGURATION. THE TABLE LISTS THE DEPLOYMENT SETUP USED IN ALL ROBOT EXPERIMENTS.

Item	Configuration
Robot	Dual-arm AgileX PiPER platform with an absolute end-effector command interface.
Cameras	Two wrist-mounted Intel RealSense D405 RGB cameras.
Control	Demonstrations are recorded at 10 Hz; the current closed-loop runner sends commands at 5 Hz.
Visual input	Both wrist views are center-cropped to a square and resized to 336×336 for tracking, feature extraction, and policy input.
Robot representation	Absolute EE10: 3-D translation, 6-D rotation representation, and one gripper value.
Runtime interface	Synchronized RGB frames, camera intrinsics, and robot state are received through ROS.

The runtime subscribes to synchronized RGB frames, camera intrinsics, and Piper state through ROS. Before actuation, it validates image freshness, inter-camera skew, robot-state availability, units, and the checkpoint observation contract. A real rollout is enabled only after checkpoint-only loading, memory/slot validation, a SAM3 runtime check, and a no-actuation dry run.

The structured rollout logs retain online tracking, FSAE, policy, and end-to-end latency. They do not retain a hardware identifier, gripper stroke, or calibrated reprojection error. Cross-paper runtime values are therefore treated as references rather than a hardware-controlled comparison.

APPENDIX B DATASETS AND ANNOTATION

A. FO30 Registration Data

We build FO30 to evaluate fine-grained identity perception among physically similar objects. The dataset contains 30 physical FOs from four coarse categories: 5 cup lids, 14 cans, 6 boxes, and 5 cups or mugs. Each physical object defines one FO, even when several objects share the same coarse category, shape, size, or package design. Before data capture, we assign every FO a stable index, project ID, FO Prompt, and coarse-category membership. The category membership is fixed before training and defines the Similar-FO groups used for hard-negative construction and evaluation.

Figure 4 shows the complete object set.

1) Single-object registration: We capture each FO alone so that registration does not depend on a co-occurring object or scene context. Each FO has 60 RGB images captured from different viewpoints. This gives 1,800 registration images in total. The capture order is preserved in the dataset manifest.

2) Identity and mask annotation: Every registration record contains the RGB image, stable FO index, project ID, FO Prompt, binary target mask, and source provenance. The mask identifies only the registered physical object. Registration masks are used to construct training scenes and supervise the predicted mask. They are never provided as prompts during deployment.

3) Per-FO memory unit: All registration views with the same FO index belong to one identity. FO Memory Learning uses these views to learn one reusable entry of FO Memory Tokens. Different objects in the same coarse category keep separate indices and separate memory entries.

Table XI summarizes the registration set.

TABLE XI
FO30 REGISTRATION COMPOSITION. SIMILAR-OBJECT GROUPS ARE FIXED BEFORE TRAINING AND DEFINE HARD-NEGATIVE CANDIDATES.

Coarse category / similar group	FOs	Registration images
Cup lids	5	300
Cans	14	840
Boxes	6	360
Cups and mugs	5	300
Total	30	1,800

4) Training-query construction: Training composites are generated only from registration RGB/mask pairs. A target-present query contains the queried FO and uses its mask and identity as supervision. A target-absent Similar-FO query removes the

queried FO but retains an object from the same fixed coarse group. This construction teaches localization and rejection without using any held-out test pixels.

The dataset manifest fixes the FO index, prompt, capture order, mask, and source path for every record. Test RGB images and masks are never decoded as training pixels. Acquisition-camera metadata, annotator count, and an independent inter-annotator mask audit are not available in the current manifest. We therefore do not use these unavailable fields to form claims or evaluation strata.



Fig. 4. **Physical objects in FO30.** The benchmark contains 30 FOs from four coarse categories. Objects within the same category provide visually similar identities for registration and held-out testing.

B. FO30 Test Query Construction

FO30-S and FO30-C are held out from all registration images and online training composites. The test scenes contain multiple objects rather than one isolated registration object. Every visible FO is associated with its stable identity and GT mask.

1) Standard and challenge scenes: FO30-S contains 600 standard test images. It has an average of 2.11 FOs per image, and 42.8% of its scenes are Similar-FO dense. FO30-C contains 400 harder images selected to increase object density and Similar-FO co-occurrence. It has an average of 2.98 FOs per image, and 80.3% of its scenes are Similar-FO dense.

2) Exhaustive image–query pairing: Each test image is paired with all 30 FO Prompts. FO30-S therefore contains 18,000 image–query pairs, while FO30-C contains 12,000 pairs. A pair is target-present when the queried FO has a GT mask in that image and target-absent otherwise. An absent pair is a *Similar-FO negative* when at least one visible FO belongs to the same fixed coarse group as the query. All other absent pairs are non-similar negatives. The same exhaustive matrix evaluates target localization, Similar-FO ranking, and absent-query rejection without sampling easier negatives.

3) Split isolation: Neither test split contributes RGB pixels, masks, checkpoint selection, or threshold tuning to FO Memory Learning. Table XVI reports the exact present and absent pair counts.

C. Robot Demonstrations and Visual Cache

Each teleoperated frame contains two wrist RGB observations, robot state, absolute action, and timestamps. The processed source datasets for Collect Can, Push Box, and Fit Lid contain 141, 120, and 120 episodes, respectively. We use a fixed budget of 120 demonstrations per task in the reported policy comparisons. Converted LeRobot episodes use fixed camera keys, a 10 Hz frame index, and absolute EE10 state/action. Source and project manifests preserve the raw HDF5 mapping, preprocessing, ordered FO slots, memory IDs, SAM3 checkpoint, and annotation status.

Annotation starts from an FO Prompt, retrieved FO Memory Tokens, or manual positive/ negative points and boxes. The frozen 336-pixel tracker propagates each seed forward and backward. Corrections address drift, under-segmentation, or over-segmentation, but modify only the current video track and never the persistent FO memory bank. Splits must be made by source sequence to prevent adjacent-frame leakage.

For every camera and frame, the cache record in Eq. (26) is

$$\left(F_t, \{m_t^k, b_t^k, v_t^k\}_{k=1}^K\right), \quad (26)$$

where $F_t \in \mathbb{R}^{256 \times 24 \times 24}$ is from frozen SAM3 FPN2, $m_t^k \in \mathbb{R}^{1 \times 24 \times 24}$, $b_t^k \in [0, 1]^4$ is normalized $[c_x, c_y, w, h]$, and v_t^k is the slot-valid flag. Missing objects retain their semantic slot but use zero masks/boxes and $v_t^k = \text{false}$. The cache manifest records checkpoint hash, camera, episode/frame, ordered slot keys, and source provenance.

APPENDIX C

PERCEPTION COMPARISON METHODS

The five reported methods differ only in how they construct the prompt representation supplied to frozen SAM3. They share the same candidate generation, detection thresholds, losses where training is applicable, test images, and evaluator. Their paper-facing names expose the role of each comparison rather than repository class names.

Table XII lists the five prompt states.

TABLE XII

PROMPT-STATE COMPARISON IN THE FO30 EXPERIMENTS. PARAMETER COUNTS ARE TRAINABLE PARAMETERS FOR ONE FO; VALID PROMPTS CONTAIN 4–8 TOKENS IN THE CURRENT BANK.

Method	Prompt construction	Deployment state	Trainable params/FO
SAM3 Text Prompt [5]	Original frozen SAM3 text-prompt output; no learned parameters.	Original prompt tokens.	0
Text-Token Optimization [15]	Direct optimization of the K valid 256-D text tokens for each FO; padding and geometry tokens stay frozen.	Tuned text tokens.	1,024–2,048
Internal Prompt Optimization [16]	Eight learned 256-D decoder prompt tokens used together with the frozen SAM3 text condition.	Decoder prompt tokens.	2,048
Concept Embedding Search [17]	Zero-initialized additive residual on the valid post-text-encoder prompt slots.	Folded refined text tokens.	1,024–2,048
FO Memory Learning (Ours)	Valid-token residual followed by an identity-initialized affine map; both are internal parameterization choices of FO Memory Learning.	Static FO Memory Tokens.	66,816–67,840

Text-Token Optimization tests whether directly optimizing the effective semantic tokens is sufficient. Internal Prompt Optimization keeps those semantic tokens unchanged and injects eight randomly initialized prompt tokens once at the decoder entrance; they participate in decoder attention but are removed before the box, mask, and presence heads, leaving the 200 detection candidates unchanged. Concept Embedding Search tests an anchored additive search in SAM3’s post-encoder prompt space. The internal-prompt and concept-search rows transfer the central prompt-learning ideas of the cited methods into our single-FO training and evaluation framework; they are not exact reproductions of the papers’ original few-shot pipelines. FO Memory Learning retains the coarse anchor, optimizes the valid tokens through its residual and affine parameterization, and folds the result into the same native prompt layout as static FO Memory Tokens.

All four trainable methods use the same 30-FO registration data, online-320 scene schedule, loss terms, validation cadence, checkpoint selection, and maximum budget of 30 epochs or 1,200 optimizer steps. Text-Token Optimization, Concept Embedding Search, and FO Memory Learning use a token learning rate of 10^{-3} ; Internal Prompt Optimization uses 0.05 for its eight decoder tokens. SAM3 weights remain frozen in every case.

APPENDIX D

FO MEMORY LEARNING IMPLEMENTATION

A. Valid-Token Parameterization

For FO i , frozen SAM3 produces $C_i \in \mathbb{R}^{32 \times 256}$. Only its non-padding positions V_i are transformed:

$$\tilde{C}_{i,V_i} = W_i (C_{i,V_i} + \Delta_i) + b_i, \quad (27)$$

where Δ_i and b_i are zero-initialized and W_i is identity-initialized. Other FO rows, padding slots, and the frozen geometry token remain unchanged. The trainable parameter count is $|V_i| \times 256 + 256 \times 256 + 256$, or 66,816–67,840 for the current 4–8 valid-token prompts.

Table XIII summarizes the implementation and training protocol.

TABLE XIII
FO MEMORY LEARNING IMPLEMENTATION. THE TABLE SUMMARIZES THE TRAINING PROTOCOL AND FROZEN COMPONENTS.

Component	Setting
Frozen modules	SAM3 image encoder, text encoder, fusion encoder, mask/box decoder, and native presence head.
Prompt layout	32 text slots of dimension 256 plus one frozen geometry token; only target-FO valid text slots are changed.
Trainable state	Valid-token residual Δ_i , affine weight W_i , and bias b_i ; these are internal to FO Memory Learning.
Online epoch	320 scenes: 256 target-present and 64 target-absent scenes containing a hard similar object; 2–4 objects per scene.
Budget	Effective batch 8, 40 optimizer steps/epoch, at most 30 epochs or 1,200 steps.
Optimizer	AdamW; residual LR 10^{-3} , affine LR 10^{-4} , weight decay 0.1, gradient clip 1.0, bfloat16.
Schedule	One inverse-square-root schedule with 20-step warmup, 20-step timescale, and 20-step cooldown.
Validation	Fresh 320-scene validation every 5 epochs; early stopping from epoch 10 with patience 2 and minimum improvement 0.005.
Selection	Mean of Joint-pass, Rank@1, 1–raw Similar-FO FPR, 1–Absent FPR, and positive F1; oracle-IoU guard 0.95.

Positive candidates use SAM3’s official Hungarian matcher with class, box-L1, and GIoU costs 2, 5, and 2, and focal parameters $\alpha = 0.25, \gamma = 2$. Positive mask supervision combines 0.8 Dice and 0.2 BCE; box-L1 and GIoU weights are both 0.1. Positive and hard-negative presence means are 1.0 and 0.1. The candidate-to- presence gradient is detached, while the native SAM3 presence head itself stays frozen.

Every online epoch is regenerated from registration RGB/mask pairs under a fixed random contract shared by all methods. The 256 positive scenes contain the target, a configured similar FO, and, when at least three objects are sampled, an ordinary non-similar distractor. The 64 hard-negative scenes remove the queried target while retaining a configured similar FO. Photometric augmentation uses mild brightness, contrast, or gamma variation; blur, noise, or JPEG degradation is applied sparsely. Global crop and rotation are disabled so that the accepted target area and visibility constraints are controlled by the scene compositor. At least 95% of each supervised target mask remains visible, and overlap occlusion is capped at 5%. Test images and masks are explicitly forbidden as compositor material.

For each positive pair, the matcher assigns the queried FO’s GT mask and tight box to one of SAM3’s candidates. The loss combines mask Dice and BCE, object classification, box L1/GIoU, and positive presence. For a target-absent hard negative, only the native presence pathway is supervised; assigning the visible similar object as a negative mask target would incorrectly teach its geometry to the queried identity.

B. Registration, Export, and Bank Assembly

The complete registration path is: (1) assign a stable FO ID and FO Prompt; (2) acquire and annotate multi-view images; (3) cache the frozen coarse prompt anchor; (4) generate the target and hard- similar-object configuration; (5) run FO Memory Learning independently for each FO; (6) select a checkpoint by online validation; (7) fold Eq. (27) into a compact artifact; and (8) validate provenance before adding it to the bank.

The safetensors artifact stores final valid-text tokens, their positions/offsets in the 32-slot layout, text mask, frozen geometry token, and geometry mask. JSON metadata stores the method, FO index, prompt, memory ID, model resolution, storage format, and hashes of the configuration, dataset manifest, base prompt, and SAM checkpoint. Bank assembly requires exactly FO indices 0–29 and identical shared hashes. At deployment, the compact tokens are restored to the native “32 text + 1 geometry” prompt. These restored values are the FO Memory Tokens. Neither the text encoder nor the training-time residual/affine map is executed.

This export boundary is important for the method claim. The text encoder is used once to establish the frozen coarse anchor during registration, but the deployed FO Prompt is only a bank lookup. The stored tokens directly reconstruct SAM3’s native prompt sequence, so repeated inference neither re-encodes the text nor runs an adapter on SAM3.

C. Detection Score and Acceptance Rule

For one image–FO query, frozen SAM3 returns 200 raw mask/box candidates, one native presence logit, and an unfused object logit per candidate. Candidate q is ranked by

$$s_q^{\text{joint}} = \sigma(s_q^{\text{pres}}) \sigma(s_q^{\text{obj}}). \quad (28)$$

A candidate is accepted when $s_q^{\text{joint}} \geq 0.5$; its mask is binarized at 0.5, and mask NMS uses IoU 0.8. The same rule, candidate budget, thresholds, and evaluator are applied to all five perception methods. Rank@1 is computed before thresholding and NMS so that a low-confidence landing on a similar object cannot disappear from the identity-ranking diagnostic.

APPENDIX E
FSAE AND POLICY INTERFACES

FSAE is the policy-side visual encoder. It receives the frozen SAM feature map together with ordered FO masks and boxes, extracts focused spatial-appearance representations, and returns either a compact DP condition or ACT visual tokens. The ordered target/context attention, relative geometry, and residual gate described below are internal refinement operations of FSAE rather than separate top-level modules.

A. Spatial-Appearance Feature Construction

The packed cache for one camera is ordered as $[F_t, m_t^1, \dots, m_t^K, b_t^1, \dots, b_t^K]$. With $K = 2$, it has shape $[266, 24, 24]$: 256 FPN2 channels, two masks, and eight constant box planes. Slot 0 is the manipulation target and slot 1 is the context/receptacle. Their order is fixed across annotation, cache, checkpoint, and online tracking.

FSAE maps each normalized box to the FPN2 grid and applies 16×16 ROIAlign to both feature and mask. A threshold-0.3 mask gate removes in-box background. Mask-normalized pooling divides pooled features by mask occupancy, preventing small foreground regions from being diluted by zeros.

B. Ordered Feature Refinement

After focused ROI extraction and mask gating, FSAE uses the mask-pooled 256-D target as one query. The context is mask-normalized to a 4×4 grid and mapped to 16 pairs of 32-D keys/values. The query also receives the relative geometry in Eq. (29):

$$g = \left[\Delta x, \Delta y, \log \frac{w_t}{w_c}, \log \frac{h_t}{h_c}, \text{IoU}(b_t, b_c), r_{t \in c} \right], \quad (29)$$

This geometry is projected from 6-D to 32-D. As an internal FSAE refinement, one-head scaled dot-product attention produces r , and a zero-initialized $32 \rightarrow 256$ map updates only target-mask support as in Eq. (30):

$$F'_t = F_t + F_t \odot \tanh(W_r r + b_r) \odot m_t. \quad (30)$$

The context stays unchanged. Missing either ordered mask yields an exact identity mapping. This attention-and-gate refinement adds 33,344 trainable parameters without changing DP condition size or ACT token count.

SAM3 is outside the policy optimization graph: its cached FPN2 features, masks, and boxes are fixed inputs. Gradients from the action loss pass through the complete FSAE visual encoder and the action backbone, but not into SAM3. The zero-initialized residual gate therefore starts from an identity mapping and learns task-specific modulation jointly with action generation; it is not a fixed handcrafted transform.

Table XIV compares the DP and ACT interfaces.

TABLE XIV
FSAE INTERFACES FOR DP AND ACT. THE TWO INTERFACES USE THE SAME FROZEN SAM3 OUTPUTS.

Item	Ours-DP	Ours-ACT
ROI output	16 heatmaps and mask-aware SpatialSoftmax; 32-D/object plus box4	Mask-normalized 8×8 grid; 64 tokens/object, 256-D projected to 512-D
Two cameras, two slots	144-D visual condition	256 visual tokens of dimension 512
Policy input	Two EE10 states give 164-D observation condition; plus 128-D time embedding gives 292-D FiLM condition	Visual tokens plus latent and state tokens give 258 encoder tokens
FSAE internal refinement	33,344 trainable parameters	33,344 trainable parameters
Complete FSAE encoder	37,456 trainable parameters	233,664 trainable parameters
Action supervision	Diffusion denoising MSE; gradients update FSAE, SpatialSoftmax adapter, and DP	Action L1 plus VAE KL; gradients update FSAE, token projections/embeddings, and ACT
Training budget	Batch 16, 150k steps, prediction horizon 24, action horizon 16	Batch 16, 100k steps, chunk 30, temporal-ensemble coefficient 0.01
Configuration family	FOM-SAM3 + FSAE with DP	FOM-SAM3 + FSAE with ACT

Table XV lists the policy configurations.

TABLE XV
POLICY TRAINING AND INFERENCE CONFIGURATION. THE BATCH SIZE IS 16 FOR EVERY TASK AND BACKBONE.

Item	Ours-DP	Ours-ACT
Optimizer	Adam, LR 10^{-4} , betas (0.95, 0.999), weight decay 10^{-6}	AdamW, LR 10^{-5} , weight decay 10^{-4}
Schedule	Cosine decay; 500-step warmup	Constant optimizer settings
Training budget	Batch 16; 150k updates	Batch 16; 100k updates
Temporal input	Current image; EE10 at $t - 1, t$	Current image and current EE10
Action target	24 steps, $t - 1:t + 22$; execute $t:t + 15$	30 steps, $t:t + 29$; emit one action with temporal ensembling
Normalization	Visual identity; state/action min-max	Visual identity; state/action mean-standard deviation
Action model	1-D conditional U-Net, widths 512/1024/2048, kernel 5, group norm 8, scale-and-bias FiLM	Transformer width 512, 8 heads, FFN 3200, 4 encoder and 1 decoder layers
Stochastic objective	DDPM, 100 train and 100 checkpoint inference steps, squared-cosine schedule, epsilon prediction	32-D VAE latent, 4 VAE encoder layers, KL weight 10
Deployment	16-action receding-horizon chunk	Temporal-ensemble coefficient 0.01; one action/control step

The complete FSAE counts exclude the frozen SAM3 feature extractor, parameter-free ROI/mask operations, fixed positional buffers, and the DP/ACT action backbone. For DP, 37,456 consists of the 33,344-parameter internal refinement and a 4,112-parameter SpatialSoftmax heatmap projection. For ACT, 233,664 includes the internal refinement, SAM-feature normalization/projection, box projection, and camera/role embeddings. RGB and FSAE variants use identical demonstrations, state/action representations, action objectives, and training budgets. At inference, FSAE and the policy are frozen together; registering a new FO changes only the FO Memory Tokens stored in the bank.

C. Online Detection and Tracking

At episode reset, FOM-SAM3 retrieves the required FO Memory Tokens from the bank and supplies them to the frozen 504-pixel SAM3.1 multiplex detector, which must initialize every ordered slot. The detected mask and box provide a global visual identity anchor for the requested FO. After resizing, these masks seed independent 336-pixel trackers for the two wrist cameras. Each tracker memory bank uses the recent image/mask history and its previous state to propagate local motion and maintain the selected instance between detector calls.

Normal control runs the tracker, FPN2, FSAE, and DP/ACT. If a required slot is invalid, actuation stops and FOM-SAM3 repeats detection with the retrieved FO Memory Tokens before the tracker is re-initialized. The persistent FO memory bank therefore supports long-term identity retrieval and global re-localization, whereas tracker memory performs short-term temporal maintenance only. Tracker state is camera-specific, episode-local, never changes slot order, and never updates the FO Memory Tokens or bank. The formal FO30 evaluator uses score threshold 0.5, mask threshold 0.5, mask-NMS IoU 0.8, 200 raw candidates, and single-instance Top-1 output. Prompt-route latency is measured on registration image `wontae_grape_can/000001.jpg` at 504-pixel SAM3 resolution. The Text Prompt and FO Memory Token routes use the same frozen image backbone, fusion encoder, decoder, threshold, and post-processing. FO Memory Tokens are loaded before timing. Five warm-up passes precede 30 alternating measurements with CUDA synchronization. Model loading, memory-file I/O, tracker initialization, and policy inference are excluded.

APPENDIX F REAL-ROBOT EVALUATION PROTOCOL

The three tasks are Can Pick & Place, Box Push, and Part Assembly, covering grasp-and-place, planar contact, and local insertion. All policies are trained in a clean single-target setting. Evaluation uses an ID single-target condition, changes the background and introduces OOD-Distractors or OOD-Similar distractors, or evaluates within-type and cross-type newly registered FOs with compatible interaction geometry. OOD-Similar denotes unseen distractor instances that share the queried target’s coarse category.

Five target markers are fixed on the table, and every method uses the same cameras, object poses, marker order, termination rule, and ten- rollout protocol. ID contains only the demonstrated target. For both OOD-Distractors and OOD-Similar, the first five trials use distractor 1 and the last five use distractor 2. For within-type scaling, 3–4 similar FOs stay at fixed locations; the query switches from new FO 1 to new FO 2 after five trials. For cross-type scaling, one geometry-compatible object set is evaluated twice at each marker. Only the queried target moves across markers.

A rollout succeeds only if the robot first interacts with the queried FO and then completes the task; interacting with a wrong FO is always a failure. New-FO conditions add only FO Memory Tokens to the bank and do not add demonstrations

or update SAM3, FSAE, DP, or ACT. This evaluates frozen- skill reuse for compatible geometry, not zero-shot acquisition of an arbitrary new manipulation skill. A reported success requires both correct-FO interaction and task completion; wrong-FO contact, timeout, or an unfinished task is a failure. Exact task-specific pose tolerances and timeout values are not retained in the current summary records and are therefore not reconstructed post hoc.

APPENDIX G

ADDITIONAL EXPERIMENTAL RESULTS AND OBSERVATIONS

A. Detailed FO Perception Metrics

We name the 600-image standard benchmark *FO30-Standard* (FO30-S) and the 400-image similar-object-dense stress test *FO30-Challenge* (FO30-C). These paper-facing names map to the internal evaluation profiles `standard600` and `previous400`, respectively. A similar-object-dense scene contains at least two FOs from the same predefined similar group.

TABLE XVI
COMPOSITION OF THE FO30 TEST BENCHMARKS. PRESENT AND ABSENT REFER TO EXHAUSTIVE IMAGE-FO QUERY PAIRS.

Benchmark	Images	Queries	Present	Absent	FOs / image	Similar-dense scenes
FO30-S (Standard)	600	18,000	1,268	16,732	2.11	42.8%
FO30-C (Challenge)	400	12,000	1,192	10,808	2.98	80.3%

Tables XVII and XIX expand the four metrics reported in Table I without changing the protocol or evaluator. The GT-present report evaluates target localization and segmentation only on images containing the queried FO. The Full-FO report evaluates all FO-image pairs, including target-absent similar-object scenes, and therefore exposes false acceptance. All methods use the same frozen SAM3 backbone, thresholds, decoder candidate budget, and evaluator. The formal dual-report evaluator exports pmF1 and F1@0.5 rather than a standalone Dice field; we therefore do not mix the earlier small-scale Dice result with the FO30 evaluation.

B. Metric Definitions

Let a positive pair contain at least one GT instance of the queried FO. GT mIoU is averaged over GT instances after Hungarian matching; an unmatched GT contributes zero, while duplicate predictions are penalized by precision/F1 rather than counted again in mIoU. F1@0.5 uses mask IoU 0.5. pmF1 averages instance-level F1 over IoU thresholds 0.50, 0.55, ..., 0.95. Joint-pass is the fraction of matcher-assigned target candidates whose score in Eq. (28) passes 0.5. Rank@1 is the fraction for which that assigned target candidate ranks first among the 200 raw candidates.

For identity-level presence, each image-FO pair is one binary decision, yielding TP, FP, TN, FN . Absent FPR is $FP/(FP + TN)$ over all target-absent pairs; it is the quantity named `Joint_FPR` in the evaluator because acceptance uses the joint score. Similar-FO FPR restricts the denominator to absent queries for which a Similar-FO is present. Non-similar FPR uses the remaining absent pairs. IL-MCC is the Matthews correlation coefficient of the four identity-level counts. The complete grounding score in Eq. (31) is

$$\text{cgF1} = \text{pmF1} \times \text{IL-MCC}, \quad (31)$$

Thus, a method cannot score well by segmenting present objects while accepting most absent queries, or by rejecting everything while missing present targets.

TABLE XVII
GT-PRESENT LOCALIZATION AND SEGMENTATION RESULTS. ALL VALUES ARE PERCENTAGES; JOINT-PASS USES THE JOINT OBJECTNESS AND IDENTITY THRESHOLD.

Test set	Method	GT mIoU↑	pmF1↑	F1@0.5↑	Joint-pass↑	Rank@1↑
FO30-S	SAM3 Text Prompt [5]	70.18	73.57	74.00	71.85	87.85
	Text-Token Optimization [15]	84.77	81.49	82.19	86.75	90.14
	Internal Prompt Optimization [16]	68.22	73.37	73.81	69.79	87.78
	Concept Embedding Search [17]	84.85	81.53	82.31	86.91	90.54
	FO Memory Learning (Ours)	92.66	93.56	93.63	94.48	97.56
FO30-C	SAM3 Text Prompt [5]	57.46	55.89	56.23	59.06	72.48
	Text-Token Optimization [15]	68.84	62.76	63.44	70.89	75.42
	Internal Prompt Optimization [16]	53.19	56.00	56.33	54.87	71.90
	Concept Embedding Search [17]	69.63	62.95	63.69	71.73	75.17
	FO Memory Learning (Ours)	83.76	85.42	85.57	85.99	95.55

Table XVIII gives one fixed FO diagnostic.

TABLE XVIII
FRESH-VALIDATION DIAGNOSTIC FOR THE “COTTON SWAB BOX” FO. THE 320 SCENES CONTAIN 256 PRESENT AND 64 ABSENT CASES. VALUES ARE PERCENTAGES AND ARE SEPARATE FROM THE FORMAL FO30 TEST SETS.

Method	GT mIoU	F1@0.5	Joint-pass	Rank@1	Similar-FO FPR	Absent FPR
SAM3 Text Prompt [5]	0.00	0.00	0.00	55.47	35.55	0.00
Concept Embedding Search [17]	72.11	61.71	74.61	66.02	31.64	35.94
FO Memory Learning (Ours)	95.88	96.38	98.83	99.22	0.00	1.56

The original prompt can produce high-quality raw mask candidates for this object, but none of its matched candidates pass the joint score threshold; consequently, its GT mIoU and F1 are zero after the formal detection decision. The FO Memory Tokens learned by our method raise the target score while preserving mask quality, separates similar boxes, and retains low absent-query activation. The zero Absent FPR of the original prompt is not by itself a success: it results from near-universal rejection and is captured by the simultaneous zero recall/F1.

TABLE XIX
FULL-FO IDENTIFICATION AND REJECTION RESULTS. CGF1 AND FPR VALUES ARE PERCENTAGES; IL-MCC USES ITS NATIVE SCALE. THE COUNTS COVER ALL IMAGE-FO QUERY PAIRS.

Test set	Method	cgF1↑	IL-MCC↑	Joint FPR↓	Similar-FO FPR↓	Non-similar FPR↓	TP	FP	TN	FN
FO30-S	SAM3 Text Prompt [5]	24.30	0.3304	18.96	35.46	6.03	927	3,173	13,559	341
	Text-Token Optimization [15]	35.87	0.4403	17.19	34.63	3.52	1,125	2,876	13,856	143
	Internal Prompt Optimization [16]	26.30	0.3585	15.75	29.00	5.37	906	2,636	14,096	362
	Concept Embedding Search [17]	35.82	0.4394	17.26	34.80	3.52	1,125	2,888	13,844	143
	FO Memory Learning (Ours)	62.38	0.6668	6.85	13.77	1.43	1,199	1,146	15,586	69
FO30-C	SAM3 Text Prompt [5]	20.34	0.3641	14.42	35.87	3.78	743	1,558	9,250	449
	Text-Token Optimization [15]	29.41	0.4687	13.06	34.73	2.31	891	1,411	9,397	301
	Internal Prompt Optimization [16]	20.82	0.3718	12.63	31.30	3.38	711	1,365	9,443	481
	Concept Embedding Search [17]	29.77	0.4729	13.10	34.90	2.30	900	1,416	9,392	292
	FO Memory Learning (Ours)	55.98	0.6554	7.60	18.57	2.16	1,035	821	9,987	157

These tables follow the complete pipeline summary generated from the formal per-method summaries after bank-SHA verification. All five methods have both FO30-S and FO30-C results. Blue rows and bold values denote our method.

The finalized real-robot results are reported in the main paper.

C. Ablation Studies

The implementation supports the following paper-aligned ablations: (1) valid-token residual only versus residual plus affine map within FO Memory Learning; (2) removal of target-absent hard-similar scenes; (3) removal of box geometry or mask gating from FSAE; and (4) FSAE without its internal attention refinement, relative geometry, or local context features. These variants are supported by the implementation, but no matched formal results are currently available; we therefore make no numerical ablation claim. The earlier 8/16/32 independent-memory-token capacity study is not applicable to the final compact valid-token parameterization and is excluded.

D. Cross-Seed Results

FO Memory Learning and the formal FO30 comparisons use the fixed seed 7 and are not reported as cross-seed means. The real-robot entries are repeated rollouts of a trained policy, rather than independent policy- training seeds. Consequently, the current evidence supports within-protocol method comparisons but not a claim about cross-seed variance.

E. Failure Cases and Observations

We distinguish FO misidentification, detector miss, tracker drift, temporary occlusion, leaving the field of view, wrong- FO interaction, and manipulation failure after correct identification. Frames with manual correction or re-initialization must be marked separately. The current summaries do not retain audited counts by failure type or recovery status, so we report this taxonomy without claiming a quantitative trend.

APPENDIX H SUPPLEMENTARY VISUALIZATION PROTOCOL

A. Dataset Samples

Representative panels should show the multi-view registration images, target masks, FO identifiers, and data provenance. These panels expose the visual evidence used by FO Memory Learning to create each persistent entry of FO Memory Tokens rather than suggesting that registration relies on the FO name alone.

B. Target Retrieval and Mask/Box Outputs

Qualitative retrieval panels are ordered as FO Prompt, RGB observation, SAM3 Text Prompt output, FOM-SAM3 output using FO Memory Tokens, and final mask/box. Each set includes a similar-object-dense present query and an absent query so that target localization, similar-object separation, and rejection are visible in the same format.

Figure 5 compares the two prompt routes on representative scenes from both FO30 test sets.

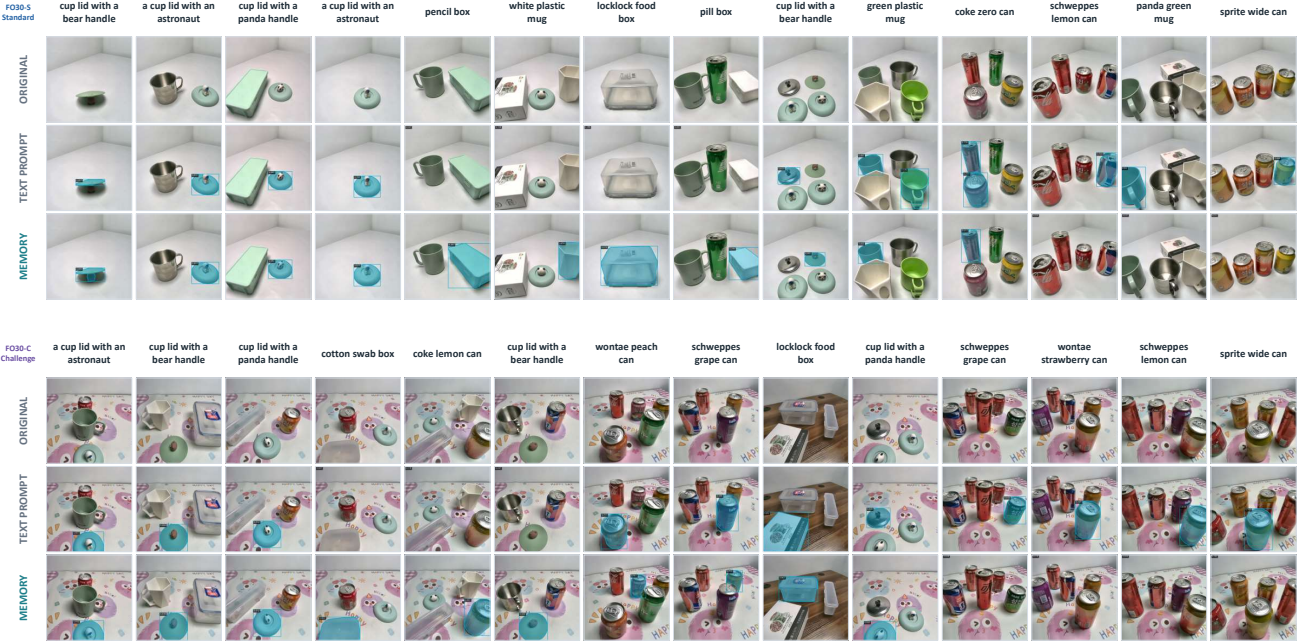


Fig. 5. **Qualitative comparison on FO30.** The top panel shows FO30-S, and the bottom panel shows FO30-C. Rows compare the original image, frozen SAM3 Text Prompt, and FOM-SAM3 with retrieved FO Memory Tokens. The examples cover target localization, Similar-FO discrimination, and absent-query rejection.

The frozen text prompt often responds to a semantically related object when the fine-grained query is ambiguous. This produces wrong-object masks in scenes with several Similar-FOs. FO Memory more often selects the queried physical identity and suppresses the other candidates. When the queried FO is absent, FOM-SAM3 returns no accepted target instead of forcing a match to the nearest Similar-FO. The same behavior is visible in the denser FO30-C scenes.

C. Continuous Tracking

Time-ordered sequences should cover identity preservation, short occlusion, target absence, tracker loss, stopped actuation, and FOM-SAM3-guided re-localization. Any manual correction or tracker re-initialization must be marked. The visualization should distinguish the initial global anchor produced by FOM-SAM3 with retrieved FO Memory Tokens from the episode-local mask propagation performed by tracker memory.

D. Representation Analysis

Representation panels may visualize FSAE’s aligned target/context ROIs, internal context attention, and target channel gate. Any dimensionality reduction must state its sample set, procedure, and random seed; it is used only for interpretation and not as evaluation evidence.

E. Real-Robot Execution

Time-ordered execution panels cover memory retrieval, approach, contact, task execution, and termination for all three tasks. The supplementary video additionally demonstrates same-scene query switching, short occlusion and recovery, re-localization after target loss, and reuse of an unchanged geometry-compatible policy after a new FO is registered. These clips are qualitative demonstrations and do not replace the rollout counts in the main paper.